Supplementary material
Accounting for seasonal variation of nasal gene expression improves transcriptome analysis of asthma

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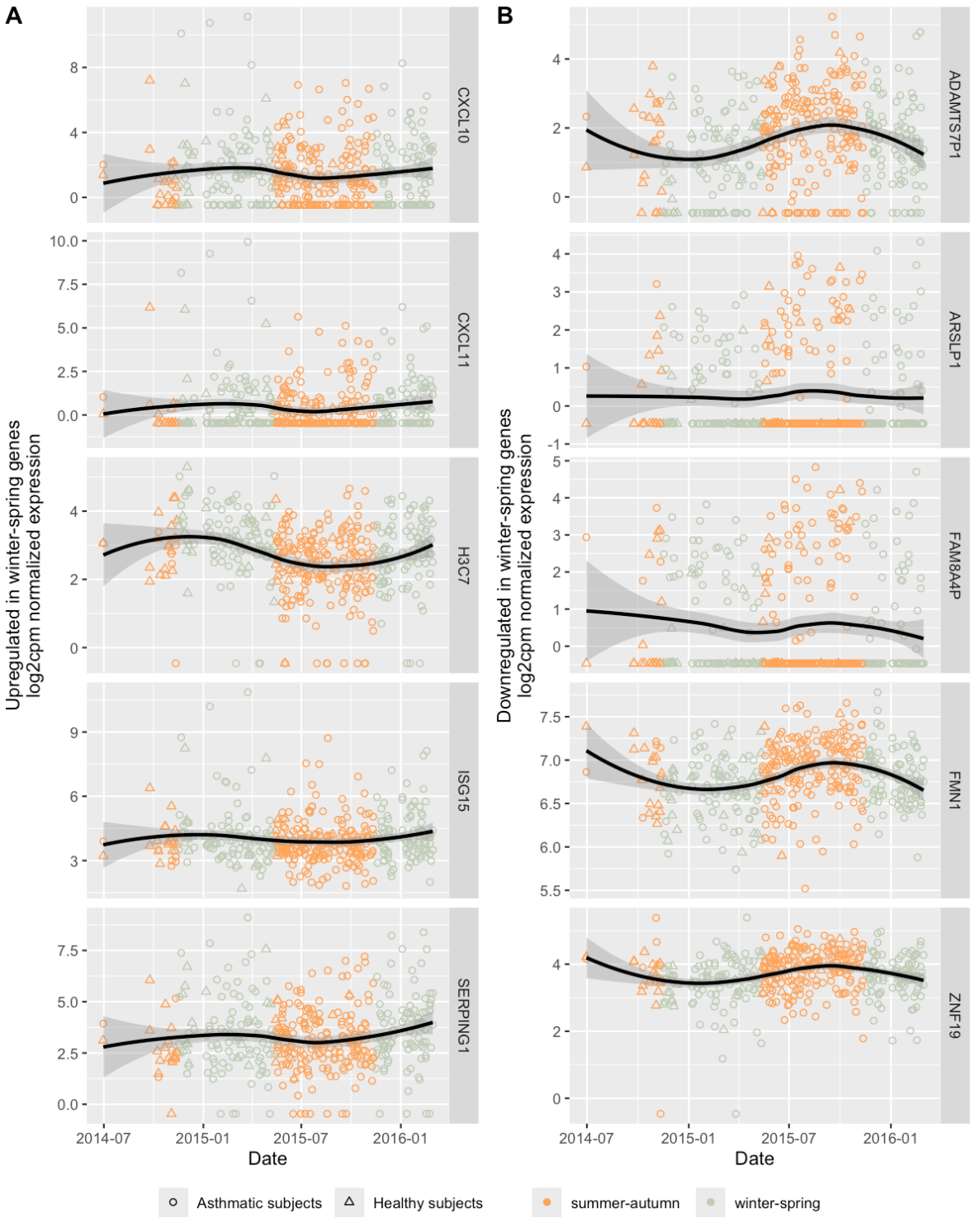
On behalf of ATLANTIS investigators
*both senior authors contributed equally

Supplementary Methods

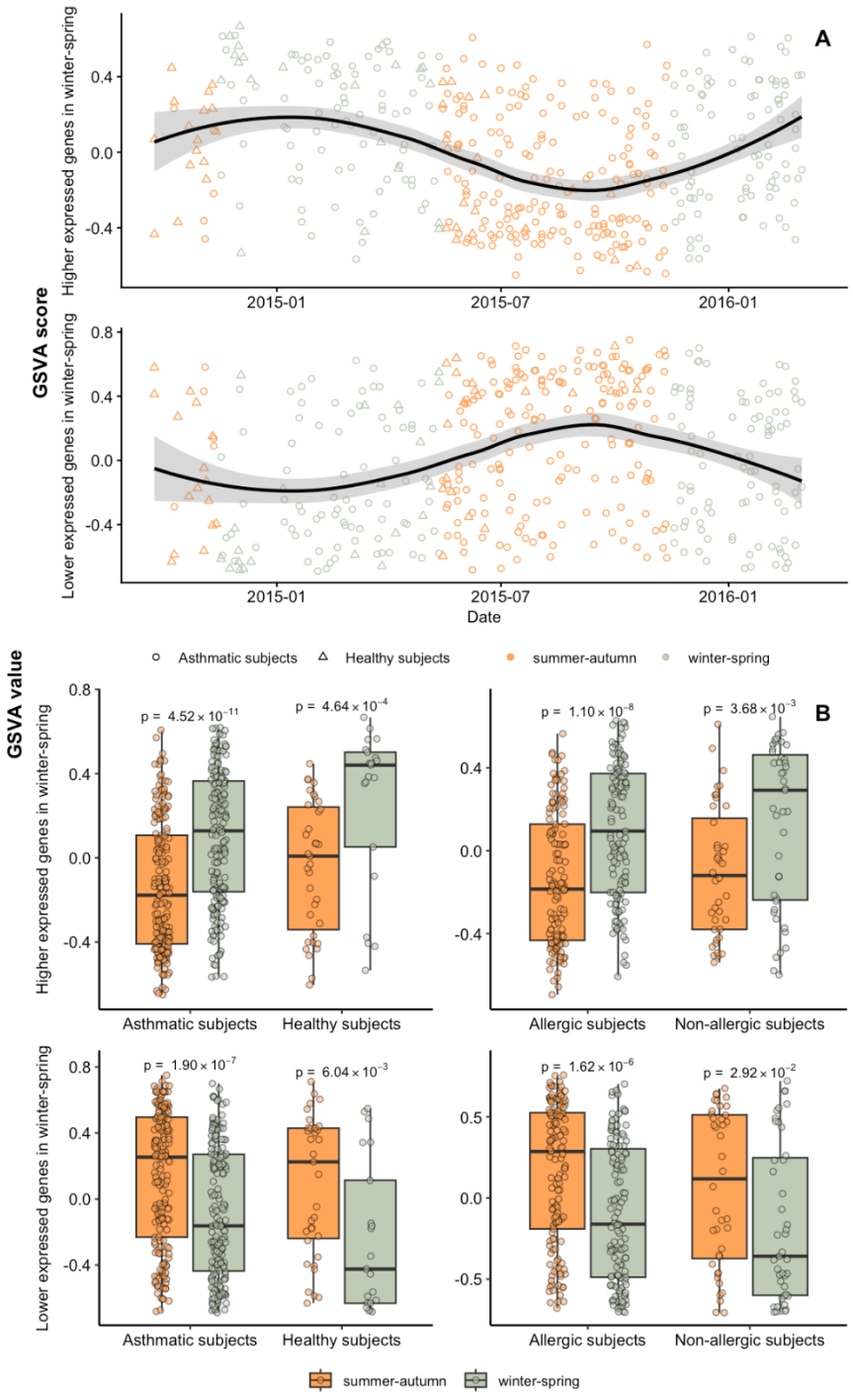
To show that our findings are not due to random fluctuations but rather a stable characteristic of the cohort we performed a permutation test by randomly swapping the season labels to subjects and performing the DGE analysis 100 times with the sex, age, smoking and asthma status as confounders. The number of differentially expressed genes with FDR <0.05 was recorded and a histogram was plotted.

Using this approach, none of the analyses showed more than 100 differentially expressed (DE) genes, in contrast to the 1731 DE genes observed when proper sampling dates were used (Supplementary Figure 4). These results confirm the robustness of our findings.

Supplementary Figures

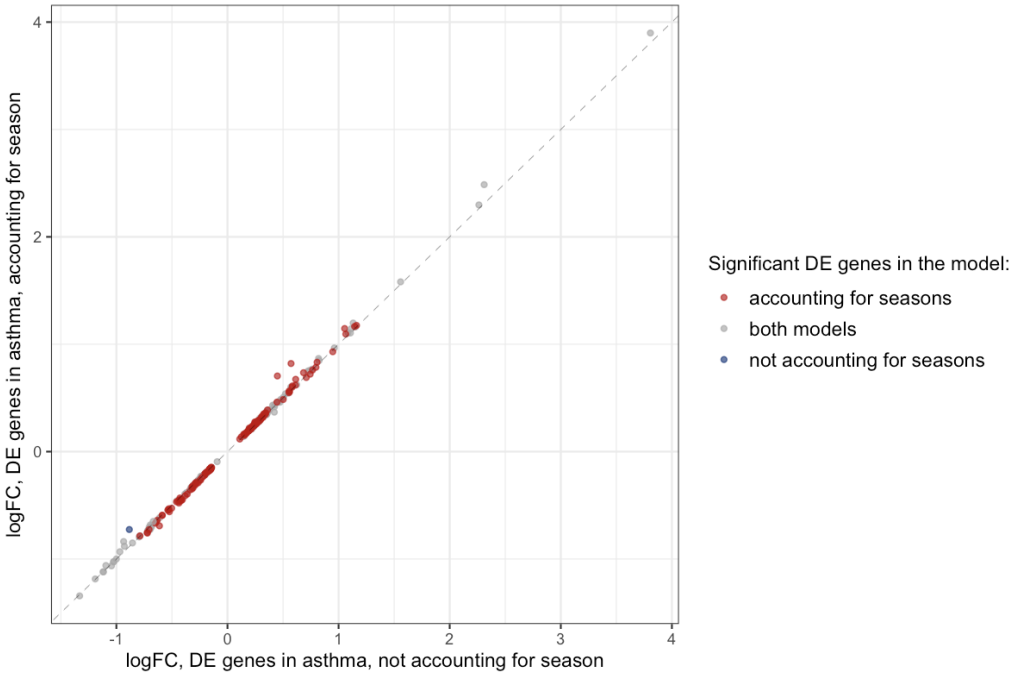


Supplementary Figure 1. *Normalized gene expression level (log cpm) over time.*
Each dot represents an individual sample, circle - asthma patient, triangle - healthy control; 2 colours represent 2 seasons: winter-spring and summer-autumn. A: top 5 genes higher expressed in winter-spring, B: top 5 genes lower expressed in winter-spring season. The line represents a locally weighted regression that captures trends by considering nearby data points within a sliding window, grey area represents 95% confidence intervals, created with geom_smooth function with "loess" method.

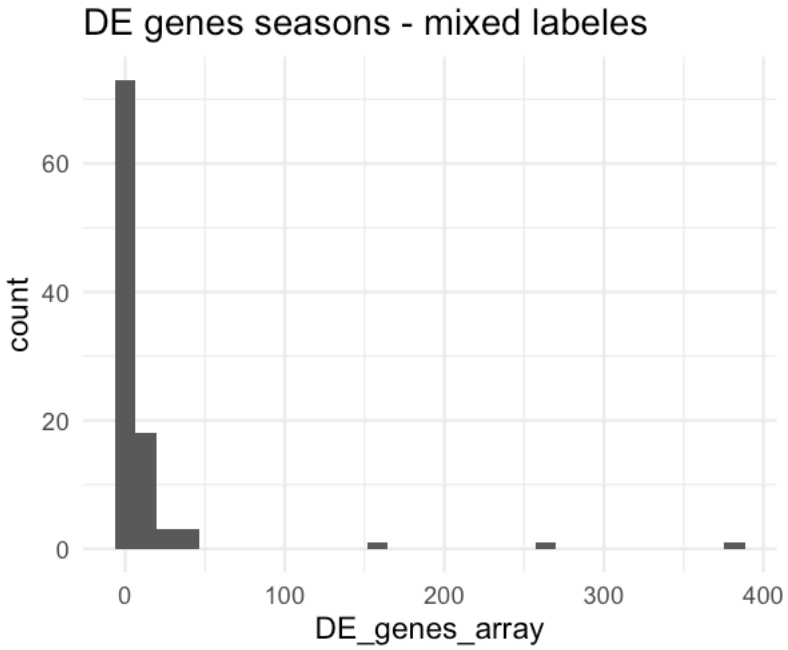


Supplementary Figure 2. *A) GSVA score per ATLANTIS sample ordered by the sample collection date for genes higher (top) and lower (bottom) expressed in winter-spring compared to summer-autumn after removal of the two samples from the earliest collection time point. The line represents*

a locally weighted regression that captures trends by considering nearby data points within a sliding window, grey area represents 95% confidence intervals. ***B)*** *GSVA scores comparison of genes with expression affected by season, analyzed separately for asthma patients and healthy participants; and allergic and non-allergic participants from the ATLANTIS cohort after removal of the two samples from the earliest collection time point. P-values from the Mann-Whitney U test are shown at the top of the plots.*



Supplementary Figure 3. *Effect of adjusting for season on asthma-associated differentially expressed genes. Scatter plot comparing log2 fold-changes (log2FC) for asthma-associated genes estimated from a model not accounting for season (x-axis) versus a model accounting for season (y-axis). Only genes reaching significance (FDR < 0.05) in at least one of the two models are shown. Colors represent genes, depending on in which model(s) they were identified as significantly differentially expressed. The dashed diagonal line indicates equality between the two log2FC estimates.*



Supplementary Figure 4. *Distribution of the number of Differentially expressed genes for 100 permutation tests with seasonal label-shuffling.*

Supplementary Tables

	Summer-autumn	**Winter-spring**	p-value
N subjects	112	213	
Male, n (%)	53 (47.3)	88 (41.3)	0.357
Age, years	16		
Current Smoker, n (%)	9 (8.0)	15 (7.0)	0.919
Asthmatic subjects, n (%)	6 (5.4)	18 (8.5)	0.429
Inhaled corticosteroids use, n (%)	6 (5.4)	12 (5.6)	1.000

Supplementary Table 1. *Characteristics of the participants in the replication cohort PIAMA. All participants were 16 years old. Chi-square test was used for categorical data.*

Pathways enriched in genes higher expressed in winter-spring season								
ONTOLOGY	ID	Description	setSize	enrichmentScore	NES	pvalue	p.adjust	qvalue
BP	GO:0051873	killing by host of symbiont cells	10	0.793	1.68	0.00313	0.0151	0.00978
BP	GO:0040019	positive regulation of embryonic development	10	0.762	1.61	0.00755	0.0311	0.0202
BP	GO:2000310	regulation of NMDA receptor activity	10	0.754	1.6	0.0102	0.0398	0.0258
BP	GO:1900025	negative regulation of substrate adhesion-dependent cell spreading	11	0.814	1.75	0.000545	0.00342	0.00221
BP	GO:0046794	transport of virus	12	0.781	1.68	0.00137	0.0075	0.00486
BP	GO:0040020	regulation of meiotic nuclear division	12	0.744	1.6	0.00502	0.0224	0.0145
BP	GO:0021696	cerebellar cortex morphogenesis	12	0.725	1.56	0.00919	0.0366	0.0237
BP	GO:0051443	positive regulation of ubiquitin-protein transferase activity	13	0.851	1.85	1.72e-05	0.000165	0.000107
BP	GO:0006735	NADH regeneration	13	0.771	1.67	0.0011	0.00617	0.004
BP	GO:0061718	glucose catabolic process to pyruvate	13	0.771	1.67	0.0011	0.00617	0.004
BP	GO:0046519	sphingoid metabolic process	13	0.747	1.62	0.0029	0.0142	0.0092
BP	GO:0021602	cranial nerve morphogenesis	13	0.735	1.6	0.00472	0.0213	0.0138
BP	GO:0046653	tetrahydrofolate metabolic process	14	0.71	1.55	0.0107	0.0411	0.0267
BP	GO:2000773	negative regulation of cellular senescence	16	0.7	1.55	0.00666	0.0281	0.0182
BP	GO:0007019	microtubule depolymerization	16	0.698	1.55	0.0068	0.0286	0.0185
BP	GO:0034311	diol metabolic process	17	0.679	1.52	0.0134	0.0495	0.0321
BP	GO:0006775	fat-soluble vitamin metabolic process	19	0.685	1.54	0.00495	0.0222	0.0144
BP	GO:0039529	RIG-I signaling pathway	19	0.683	1.54	0.00517	0.0229	0.0148
BP	GO:0002418	immune response to tumor cell	20	0.69	1.56	0.00405	0.0187	0.0121
BP	GO:0003254	regulation of membrane depolarization	20	0.649	1.47	0.011	0.042	0.0272
BP	GO:0046839	phospholipid dephosphorylation	20	0.647	1.47	0.0119	0.045	0.0292
BP	GO:0033280	response to vitamin D	22	0.659	1.51	0.00949	0.0376	0.0244
BP	GO:0031670	cellular response to nutrient	25	0.631	1.46	0.012	0.0453	0.0294
BP	GO:0051293	establishment of spindle localization	29	0.704	1.64	0.00018	0.00135	0.000873
BP	GO:0051445	regulation of meiotic cell cycle	29	0.685	1.6	0.000412	0.0027	0.00175
BP	GO:0019233	sensory perception of pain	29	0.67	1.56	0.000838	0.00494	0.0032

BP	GO:0071384	cellular response to corticosteroid stimulus	29	0.635	1.48	0.00472	0.0213	0.0138
BP	GO:0072583	clathrin-dependent endocytosis	30	0.619	1.45	0.00703	0.0293	0.019
BP	GO:0033555	multicellular organismal response to stress	32	0.624	1.47	0.00435	0.0199	0.0129
BP	GO:0060135	maternal process involved in female pregnancy	33	0.658	1.55	0.000499	0.00321	0.00208
BP	GO:0061045	negative regulation of wound healing	33	0.587	1.38	0.012	0.0453	0.0294
BP	GO:0031640	killing of cells of another organism	36	0.758	1.8	8.86e-08	1.25e-06	8.13e-07
BP	GO:0141061	disruption of cell in another organism	36	0.758	1.8	8.86e-08	1.25e-06	8.13e-07
BP	GO:0019217	regulation of fatty acid metabolic process	36	0.62	1.47	0.00191	0.00993	0.00644
BP	GO:0043470	regulation of carbohydrate catabolic process	36	0.606	1.44	0.0035	0.0165	0.0107
BP	GO:0046824	positive regulation of nucleocytoplasmic transport	37	0.622	1.48	0.00161	0.0086	0.00557
BP	GO:0046847	filopodium assembly	37	0.605	1.44	0.00361	0.017	0.011
BP	GO:0141060	disruption of anatomical structure in another organism	38	0.754	1.79	2.73e-07	3.64e-06	2.36e-06
BP	GO:0051784	negative regulation of nuclear division	40	0.801	1.9	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010965	regulation of mitotic sister chromatid separation	40	0.8	1.9	1,00E-10	2.3e-09	1.49e-09
BP	GO:0046209	nitric oxide metabolic process	42	0.584	1.39	0.00859	0.0347	0.0225
BP	GO:0051492	regulation of stress fiber assembly	42	0.574	1.37	0.0116	0.044	0.0285
BP	GO:0007589	body fluid secretion	44	0.593	1.42	0.00372	0.0174	0.0113
BP	GO:0110020	regulation of actomyosin structure organization	45	0.566	1.35	0.0125	0.0466	0.0302
BP	GO:0030510	regulation of BMP signaling pathway	47	0.586	1.41	0.00368	0.0172	0.0112
BP	GO:1905818	regulation of chromosome separation	50	0.772	1.86	1,00E-10	2.3e-09	1.49e-09
BP	GO:0060761	negative regulation of response to cytokine stimulus	50	0.681	1.64	3.63e-06	4.02e-05	2.61e-05
BP	GO:0006096	glycolytic process	51	0.682	1.64	2.53e-06	2.88e-05	1.87e-05
BP	GO:0051304	chromosome separation	53	0.765	1.84	1,00E-10	2.3e-09	1.49e-09
BP	GO:0030038	contractile actin filament bundle assembly	53	0.591	1.42	0.00167	0.00885	0.00574
BP	GO:0043149	stress fiber assembly	53	0.591	1.42	0.00167	0.00885	0.00574
BP	GO:0010507	negative regulation of autophagy	53	0.567	1.36	0.00703	0.0293	0.019
BP	GO:0051851	modulation by host of symbiont process	55	0.546	1.32	0.0123	0.0462	0.0299
BP	GO:0140747	regulation of ncRNA transcription	57	0.557	1.35	0.00949	0.0376	0.0244

BP	GO:0001910	regulation of leukocyte mediated cytotoxicity	62	0.585	1.42	0.0011	0.00617	0.004
BP	GO:0140888	interferon-mediated signaling pathway	64	0.674	1.64	1.95e-07	2.65e-06	1.72e-06
BP	GO:0090398	cellular senescence	64	0.546	1.33	0.00576	0.0249	0.0162
BP	GO:0006090	pyruvate metabolic process	65	0.648	1.58	5.84e-06	6.21e-05	4.03e-05
BP	GO:0002709	regulation of T cell mediated immunity	65	0.597	1.46	0.000371	0.00247	0.0016
BP	GO:0034446	substrate adhesion-dependent cell spreading	65	0.581	1.42	0.000948	0.00546	0.00354
BP	GO:0050868	negative regulation of T cell activation	72	0.601	1.48	6.04e-05	0.000505	0.000328
BP	GO:0071621	granulocyte chemotaxis	72	0.594	1.46	0.000112	0.00088	0.00057
BP	GO:0007088	regulation of mitotic nuclear division	73	0.732	1.8	1,00E-10	2.3e-09	1.49e-09
BP	GO:0097553	calcium ion transmembrane import into cytosol	74	0.617	1.52	2.19e-05	0.000205	0.000133
BP	GO:0006334	nucleosome assembly	75	0.72	1.77	1,00E-10	2.3e-09	1.49e-09
BP	GO:0045727	positive regulation of translation	76	0.559	1.38	0.00111	0.00626	0.00406
BP	GO:0048659	smooth muscle cell proliferation	76	0.519	1.28	0.0126	0.0471	0.0305
BP	GO:0007584	response to nutrient	78	0.631	1.56	8.39e-07	1.03e-05	6.7e-06
BP	GO:0006195	purine nucleotide catabolic process	79	0.639	1.58	4.22e-07	5.42e-06	3.52e-06
BP	GO:0002367	cytokine production involved in immune response	80	0.549	1.35	0.00143	0.00774	0.00502
BP	GO:0050921	positive regulation of chemotaxis	80	0.533	1.32	0.00442	0.0202	0.0131
BP	GO:0051783	regulation of nuclear division	81	0.709	1.75	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051983	regulation of chromosome segregation	82	0.727	1.8	1,00E-10	2.3e-09	1.49e-09
BP	GO:0007093	mitotic cell cycle checkpoint signaling	82	0.701	1.73	1,00E-10	2.3e-09	1.49e-09
BP	GO:2000241	regulation of reproductive process	82	0.599	1.48	2.95e-05	0.000268	0.000173
BP	GO:0030216	keratinocyte differentiation	89	0.521	1.3	0.00755	0.0311	0.0202
BP	GO:0034728	nucleosome organization	90	0.709	1.77	1,00E-10	2.3e-09	1.49e-09
BP	GO:0001890	placenta development	91	0.609	1.52	6.74e-06	7.07e-05	4.59e-05
BP	GO:0044706	multi-multicellular organism process	91	0.561	1.4	0.000527	0.00333	0.00216
BP	GO:0001909	leukocyte mediated cytotoxicity	92	0.569	1.42	0.000185	0.00137	0.00089
BP	GO:0006606	protein import into nucleus	92	0.549	1.37	0.000911	0.00529	0.00343
BP	GO:0046718	symbiont entry into host cell	94	0.556	1.39	0.000362	0.00242	0.00157
BP	GO:0009612	response to mechanical stimulus	94	0.523	1.31	0.00398	0.0184	0.0119

BP	GO:0051170	import into nucleus	97	0.539	1.35	0.00082	0.00486	0.00315
BP	GO:1903900	regulation of viral life cycle	99	0.633	1.59	3.94e-08	6,00E-07	3.89e-07
BP	GO:0006006	glucose metabolic process	101	0.494	1.24	0.0135	0.0497	0.0322
BP	GO:0007018	microtubule-based movement	102	0.533	1.34	0.00154	0.00825	0.00535
BP	GO:0007596	blood coagulation	110	0.495	1.24	0.0135	0.0497	0.0322
BP	GO:0002700	regulation of production of molecular mediator of immune response	111	0.532	1.34	0.000783	0.00466	0.00302
BP	GO:0000075	cell cycle checkpoint signaling	112	0.679	1.71	1,00E-10	2.3e-09	1.49e-09
BP	GO:1904062	regulation of monoatomic cation transmembrane transport	112	0.568	1.43	2.58e-05	0.000236	0.000153
BP	GO:0006959	humoral immune response	120	0.655	1.65	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031334	positive regulation of protein-containing complex assembly	121	0.502	1.27	0.00383	0.0178	0.0115
BP	GO:0046434	organophosphate catabolic process	123	0.543	1.37	0.000244	0.00173	0.00112
BP	GO:0051321	meiotic cell cycle	124	0.632	1.6	2.51e-09	4.66e-08	3.02e-08
BP	GO:0006874	intracellular calcium ion homeostasis	124	0.59	1.49	1.48e-06	1.76e-05	1.14e-05
BP	GO:0051222	positive regulation of protein transport	125	0.537	1.36	0.000158	0.00119	0.000774
BP	GO:0000070	mitotic sister chromatid segregation	126	0.696	1.76	1,00E-10	2.3e-09	1.49e-09
BP	GO:0042098	T cell proliferation	130	0.531	1.34	0.000399	0.00262	0.0017
BP	GO:0022412	cellular process involved in reproduction in multicellular organism	130	0.524	1.33	0.000746	0.00447	0.0029
BP	GO:0055074	calcium ion homeostasis	135	0.589	1.49	2.22e-07	3,00E-06	1.94e-06
BP	GO:0000819	sister chromatid segregation	148	0.678	1.72	1,00E-10	2.3e-09	1.49e-09
BP	GO:0001666	response to hypoxia	152	0.487	1.24	0.00383	0.0178	0.0115
BP	GO:0141091	transforming growth factor beta receptor superfamily signaling pathway	161	0.487	1.24	0.00309	0.0149	0.00969
BP	GO:0050870	positive regulation of T cell activation	162	0.536	1.37	1.97e-05	0.000186	0.000121
BP	GO:0009306	protein secretion	162	0.529	1.35	4.91e-05	0.000425	0.000275
BP	GO:0051235	maintenance of location	165	0.566	1.44	2.86e-07	3.78e-06	2.45e-06
BP	GO:0010948	negative regulation of cell cycle process	169	0.609	1.56	1,00E-10	2.3e-09	1.49e-09
BP	GO:0140014	mitotic nuclear division	176	0.67	1.72	1,00E-10	2.3e-09	1.49e-09
BP	GO:0045787	positive regulation of cell cycle	184	0.606	1.55	1,00E-10	2.3e-09	1.49e-09

BP	GO:0002449	lymphocyte mediated immunity	185	0.588	1.51	5.98e-10	1.23e-08	7.95e-09
BP	GO:0098813	nuclear chromosome segregation	188	0.661	1.69	1,00E-10	2.3e-09	1.49e-09
BP	GO:0099177	regulation of trans-synaptic signaling	193	0.465	1.19	0.0114	0.0436	0.0283
BP	GO:1901990	regulation of mitotic cell cycle phase transition	197	0.628	1.61	1,00E-10	2.3e-09	1.49e-09
BP	GO:0019058	viral life cycle	201	0.589	1.52	1,00E-10	2.3e-09	1.49e-09
BP	GO:0140694	non-membrane-bounded organelle assembly	216	0.535	1.38	1.29e-06	1.55e-05	1.01e-05
BP	GO:0007610	behavior	219	0.468	1.21	0.00561	0.0244	0.0158
BP	GO:0045786	negative regulation of cell cycle	221	0.579	1.49	1,00E-10	2.3e-09	1.49e-09
BP	GO:0000280	nuclear division	238	0.652	1.69	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009615	response to virus	242	0.588	1.52	1,00E-10	2.3e-09	1.49e-09
BP	GO:1901987	regulation of cell cycle phase transition	250	0.6	1.55	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044772	mitotic cell cycle phase transition	252	0.62	1.6	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006935	chemotaxis	252	0.465	1.2	0.00235	0.0119	0.00771
BP	GO:0042330	taxis	252	0.465	1.2	0.00235	0.0119	0.00771
BP	GO:0019693	ribose phosphate metabolic process	253	0.472	1.22	0.00126	0.00698	0.00453
BP	GO:0007059	chromosome segregation	261	0.625	1.62	1,00E-10	2.3e-09	1.49e-09
BP	GO:0000226	microtubule cytoskeleton organization	262	0.588	1.52	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044283	small molecule biosynthetic process	263	0.45	1.17	0.0125	0.0466	0.0302
BP	GO:0048285	organelle fission	267	0.632	1.64	1,00E-10	2.3e-09	1.49e-09
BP	GO:1903829	positive regulation of protein localization	268	0.487	1.26	0.000142	0.00108	0.000702
BP	GO:0016032	viral process	273	0.581	1.5	1,00E-10	2.3e-09	1.49e-09
BP	GO:0002683	negative regulation of immune system process	280	0.573	1.48	1,00E-10	2.3e-09	1.49e-09
BP	GO:0019221	cytokine-mediated signaling pathway	280	0.569	1.48	1,00E-10	2.3e-09	1.49e-09
BP	GO:0080135	regulation of cellular response to stress	288	0.487	1.27	5.53e-05	0.000467	0.000303
BP	GO:0007346	regulation of mitotic cell cycle	291	0.589	1.53	1,00E-10	2.3e-09	1.49e-09
BP	GO:0071417	cellular response to organonitrogen compound	291	0.485	1.26	7.95e-05	0.000649	0.000421
BP	GO:0045088	regulation of innate immune response	292	0.528	1.37	4.77e-08	7.11e-07	4.61e-07
BP	GO:0032504	multicellular organism reproduction	305	0.53	1.38	2.04e-08	3.24e-07	2.1e-07
BP	GO:0044770	cell cycle phase transition	310	0.592	1.54	1,00E-10	2.3e-09	1.49e-09

BP	GO:0007017	microtubule-based process	338	0.564	1.47	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006338	chromatin remodeling	354	0.584	1.52	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051276	chromosome organization	360	0.622	1.62	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051301	cell division	360	0.601	1.57	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009617	response to bacterium	370	0.568	1.48	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044092	negative regulation of molecular function	371	0.553	1.44	1,00E-10	2.3e-09	1.49e-09
BP	GO:0043632	modification-dependent macromolecule catabolic process	383	0.455	1.19	0.000691	0.0042	0.00272
BP	GO:0010564	regulation of cell cycle process	392	0.569	1.49	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006325	chromatin organization	417	0.564	1.47	1,00E-10	2.3e-09	1.49e-09
BP	GO:0040007	growth	430	0.461	1.2	0.000194	0.00142	0.000921
BP	GO:1903047	mitotic cell cycle process	438	0.605	1.58	1,00E-10	2.3e-09	1.49e-09
BP	GO:0030155	regulation of cell adhesion	450	0.52	1.36	1,00E-10	2.3e-09	1.49e-09
BP	GO:0008284	positive regulation of cell population proliferation	465	0.549	1.44	1,00E-10	2.3e-09	1.49e-09
BP	GO:0071824	protein-DNA complex organization	467	0.549	1.44	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051248	negative regulation of protein metabolic process	470	0.538	1.41	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031347	regulation of defense response	470	0.532	1.39	1,00E-10	2.3e-09	1.49e-09
BP	GO:0071345	cellular response to cytokine stimulus	479	0.553	1.45	1,00E-10	2.3e-09	1.49e-09
BP	GO:2000145	regulation of cell motility	513	0.506	1.33	1.84e-09	3.44e-08	2.23e-08
BP	GO:0000278	mitotic cell cycle	515	0.604	1.58	1,00E-10	2.3e-09	1.49e-09
BP	GO:0060341	regulation of cellular localization	518	0.466	1.22	9.52e-06	9.68e-05	6.28e-05
BP	GO:0022414	reproductive process	531	0.542	1.42	1,00E-10	2.3e-09	1.49e-09
BP	GO:0000003	reproduction	535	0.543	1.43	1,00E-10	2.3e-09	1.49e-09
BP	GO:0034097	response to cytokine	537	0.558	1.47	1,00E-10	2.3e-09	1.49e-09
BP	GO:0030097	hemopoiesis	548	0.518	1.36	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006259	DNA metabolic process	555	0.508	1.34	1,00E-10	2.3e-09	1.49e-09
BP	GO:0050776	regulation of immune response	562	0.528	1.39	1,00E-10	2.3e-09	1.49e-09
BP	GO:0045087	innate immune response	570	0.558	1.47	1,00E-10	2.3e-09	1.49e-09
BP	GO:0045321	leukocyte activation	577	0.506	1.33	1,00E-10	2.3e-09	1.49e-09

BP	GO:0051726	regulation of cell cycle	599	0.544	1.43	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010628	positive regulation of gene expression	604	0.509	1.34	1,00E-10	2.3e-09	1.49e-09
BP	GO:0033043	regulation of organelle organization	615	0.5	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0032101	regulation of response to external stimulus	622	0.529	1.39	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051094	positive regulation of developmental process	627	0.517	1.36	1,00E-10	2.3e-09	1.49e-09
BP	GO:0040011	locomotion	630	0.489	1.29	2.93e-09	5.26e-08	3.41e-08
BP	GO:0019220	regulation of phosphate metabolic process	637	0.503	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051174	regulation of phosphorus metabolic process	638	0.503	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0140546	defense response to symbiont	644	0.559	1.47	1,00E-10	2.3e-09	1.49e-09
BP	GO:0001775	cell activation	644	0.508	1.34	1,00E-10	2.3e-09	1.49e-09
BP	GO:0002684	positive regulation of immune system process	645	0.523	1.38	1,00E-10	2.3e-09	1.49e-09
BP	GO:0007267	cell-cell signaling	661	0.462	1.22	1.57e-06	1.86e-05	1.21e-05
BP	GO:0007010	cytoskeleton organization	666	0.505	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0022402	cell cycle process	672	0.561	1.48	1,00E-10	2.3e-09	1.49e-09
BP	GO:0098542	defense response to other organism	683	0.559	1.47	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006468	protein phosphorylation	690	0.498	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031399	regulation of protein modification process	703	0.491	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:2000026	regulation of multicellular organismal development	719	0.512	1.35	1,00E-10	2.3e-09	1.49e-09
BP	GO:0007155	cell adhesion	726	0.502	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0050790	regulation of catalytic activity	753	0.529	1.4	1,00E-10	2.3e-09	1.49e-09
BP	GO:0042981	regulation of apoptotic process	756	0.498	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0016477	cell migration	761	0.499	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0080134	regulation of response to stress	771	0.519	1.37	1,00E-10	2.3e-09	1.49e-09
BP	GO:0045595	regulation of cell differentiation	772	0.505	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:1901700	response to oxygen-containing compound	786	0.488	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0043067	regulation of programmed cell death	789	0.496	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0045893	positive regulation of DNA-templated transcription	794	0.487	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048870	cell motility	806	0.502	1.33	1,00E-10	2.3e-09	1.49e-09

BP	GO:0042127	regulation of cell population proliferation	822	0.535	1.41	1,00E-10	2.3e-09	1.49e-09
BP	GO:0016310	phosphorylation	822	0.49	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051240	positive regulation of multicellular organismal process	825	0.496	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006508	proteolysis	835	0.487	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051707	response to other organism	845	0.563	1.49	1,00E-10	2.3e-09	1.49e-09
BP	GO:0043207	response to external biotic stimulus	846	0.563	1.49	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009607	response to biotic stimulus	867	0.56	1.48	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048585	negative regulation of response to stimulus	872	0.496	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0002682	regulation of immune system process	880	0.525	1.39	1,00E-10	2.3e-09	1.49e-09
BP	GO:0065003	protein-containing complex assembly	888	0.5	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009888	tissue development	912	0.482	1.28	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044419	biological process involved in interspecies interaction between organisms	939	0.556	1.47	1,00E-10	2.3e-09	1.49e-09
BP	GO:0007049	cell cycle	949	0.535	1.42	1,00E-10	2.3e-09	1.49e-09
BP	GO:1902531	regulation of intracellular signal transduction	952	0.482	1.28	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006952	defense response	955	0.553	1.46	1,00E-10	2.3e-09	1.49e-09
BP	GO:0032879	regulation of localization	959	0.491	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0141124	intracellular signaling cassette	966	0.458	1.21	1.11e-07	1.56e-06	1.01e-06
BP	GO:0071310	cellular response to organic substance	981	0.529	1.4	1,00E-10	2.3e-09	1.49e-09
BP	GO:0045935	positive regulation of nucleobase-containing compound metabolic process	985	0.478	1.27	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006955	immune response	991	0.537	1.42	1,00E-10	2.3e-09	1.49e-09
BP	GO:0008283	cell population proliferation	992	0.531	1.41	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006915	apoptotic process	1011	0.493	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0065009	regulation of molecular function	1018	0.517	1.37	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010558	negative regulation of macromolecule biosynthetic process	1052	0.493	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0012501	programmed cell death	1061	0.495	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0008219	cell death	1063	0.496	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031327	negative regulation of cellular biosynthetic process	1073	0.491	1.3	1,00E-10	2.3e-09	1.49e-09

BP	GO:0009890	negative regulation of biosynthetic process	1077	0.491	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051172	negative regulation of nitrogen compound metabolic process	1101	0.507	1.34	1,00E-10	2.3e-09	1.49e-09
BP	GO:0050793	regulation of developmental process	1173	0.503	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009653	anatomical structure morphogenesis	1183	0.467	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051246	regulation of protein metabolic process	1207	0.498	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048584	positive regulation of response to stimulus	1220	0.496	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051128	regulation of cellular component organization	1224	0.48	1.28	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048468	cell development	1246	0.487	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010557	positive regulation of macromolecule biosynthetic process	1249	0.49	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048513	animal organ development	1262	0.492	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0043933	protein-containing complex organization	1269	0.498	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031328	positive regulation of cellular biosynthetic process	1281	0.489	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0070887	cellular response to chemical stimulus	1285	0.51	1.35	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009891	positive regulation of biosynthetic process	1285	0.49	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0008104	protein localization	1288	0.472	1.25	1,00E-10	2.3e-09	1.49e-09
BP	GO:0070727	cellular macromolecule localization	1293	0.472	1.25	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010033	response to organic substance	1304	0.519	1.38	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031324	negative regulation of cellular metabolic process	1308	0.496	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006796	phosphate-containing compound metabolic process	1316	0.466	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006793	phosphorus metabolic process	1325	0.467	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0007166	cell surface receptor signaling pathway	1328	0.491	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010605	negative regulation of macromolecule metabolic process	1337	0.505	1.34	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009605	response to external stimulus	1357	0.545	1.45	1,00E-10	2.3e-09	1.49e-09
BP	GO:0035556	intracellular signal transduction	1386	0.498	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0002376	immune system process	1405	0.532	1.41	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051239	regulation of multicellular organismal process	1414	0.497	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009892	negative regulation of metabolic process	1439	0.502	1.34	1,00E-10	2.3e-09	1.49e-09

BP	GO:0006355	regulation of DNA-templated transcription	1453	0.458	1.22	1,00E-10	2.3e-09	1.49e-09
BP	GO:2001141	regulation of RNA biosynthetic process	1462	0.458	1.22	1,00E-10	2.3e-09	1.49e-09
BP	GO:0022607	cellular component assembly	1468	0.49	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051173	positive regulation of nitrogen compound metabolic process	1493	0.478	1.27	1,00E-10	2.3e-09	1.49e-09
BP	GO:0033036	macromolecule localization	1498	0.467	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009966	regulation of signal transduction	1517	0.485	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006351	DNA-templated transcription	1517	0.456	1.21	1,00E-10	2.3e-09	1.49e-09
BP	GO:0032774	RNA biosynthetic process	1534	0.456	1.21	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044085	cellular component biogenesis	1636	0.482	1.28	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031325	positive regulation of cellular metabolic process	1639	0.484	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048731	system development	1659	0.456	1.22	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010646	regulation of cell communication	1671	0.485	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0023051	regulation of signaling	1676	0.484	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0042221	response to chemical	1685	0.502	1.34	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010604	positive regulation of macromolecule metabolic process	1696	0.488	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051641	cellular localization	1697	0.477	1.27	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006996	organelle organization	1714	0.496	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0034654	nucleobase-containing compound biosynthetic process	1759	0.456	1.21	1,00E-10	2.3e-09	1.49e-09
BP	GO:0019219	regulation of nucleobase-containing compound metabolic process	1770	0.458	1.22	1,00E-10	2.3e-09	1.49e-09
BP	GO:0018130	heterocycle biosynthetic process	1796	0.453	1.21	1,00E-10	2.3e-09	1.49e-09
BP	GO:0019438	aromatic compound biosynthetic process	1799	0.456	1.22	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009893	positive regulation of metabolic process	1844	0.486	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:1901362	organic cyclic compound biosynthetic process	1853	0.454	1.21	1,00E-10	2.3e-09	1.49e-09
BP	GO:0030154	cell differentiation	1881	0.48	1.28	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048869	cellular developmental process	1882	0.479	1.28	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006950	response to stress	1977	0.519	1.38	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048583	regulation of response to stimulus	1993	0.492	1.31	1,00E-10	2.3e-09	1.49e-09

BP	GO:0007275	multicellular organism development	2006	0.483	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051234	establishment of localization	2104	0.465	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010468	regulation of gene expression	2175	0.477	1.27	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044271	cellular nitrogen compound biosynthetic process	2184	0.457	1.22	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010556	regulation of macromolecule biosynthetic process	2235	0.474	1.26	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031326	regulation of cellular biosynthetic process	2288	0.473	1.26	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009889	regulation of biosynthetic process	2299	0.473	1.26	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048523	negative regulation of cellular process	2367	0.518	1.38	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051179	localization	2445	0.478	1.28	1,00E-10	2.3e-09	1.49e-09
BP	GO:0019538	protein metabolic process	2500	0.465	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0007165	signal transduction	2517	0.494	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051171	regulation of nitrogen compound metabolic process	2529	0.485	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048519	negative regulation of biological process	2530	0.524	1.4	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048856	anatomical structure development	2533	0.482	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0080090	regulation of primary metabolic process	2583	0.483	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006139	nucleobase-containing compound metabolic process	2597	0.44	1.17	1,00E-10	2.3e-09	1.49e-09
BP	GO:0046483	heterocycle metabolic process	2673	0.44	1.18	1,00E-10	2.3e-09	1.49e-09
BP	GO:0023052	signaling	2690	0.492	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006725	cellular aromatic compound metabolic process	2693	0.441	1.18	1,00E-10	2.3e-09	1.49e-09
BP	GO:0007154	cell communication	2728	0.492	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0032502	developmental process	2738	0.482	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048522	positive regulation of cellular process	2753	0.503	1.34	1,00E-10	2.3e-09	1.49e-09
BP	GO:0031323	regulation of cellular metabolic process	2772	0.484	1.29	1,00E-10	2.3e-09	1.49e-09
BP	GO:0060255	regulation of macromolecule metabolic process	2785	0.493	1.32	1,00E-10	2.3e-09	1.49e-09
BP	GO:1901360	organic cyclic compound metabolic process	2785	0.441	1.18	1,00E-10	2.3e-09	1.49e-09
BP	GO:0010467	gene expression	2873	0.45	1.2	1,00E-10	2.3e-09	1.49e-09
BP	GO:0034641	cellular nitrogen compound metabolic process	2922	0.445	1.19	1,00E-10	2.3e-09	1.49e-09
BP	GO:1901564	organonitrogen compound metabolic process	2988	0.464	1.24	1,00E-10	2.3e-09	1.49e-09

BP	GO:0032501	multicellular organismal process	2989	0.489	1.3	1,00E-10	2.3e-09	1.49e-09
BP	GO:0048518	positive regulation of biological process	3006	0.511	1.36	1,00E-10	2.3e-09	1.49e-09
BP	GO:0019222	regulation of metabolic process	3036	0.49	1.31	1,00E-10	2.3e-09	1.49e-09
BP	GO:0016043	cellular component organization	3077	0.5	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009059	macromolecule biosynthetic process	3115	0.446	1.19	1,00E-10	2.3e-09	1.49e-09
BP	GO:0051716	cellular response to stimulus	3193	0.496	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0071840	cellular component organization or biogenesis	3208	0.496	1.33	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044249	cellular biosynthetic process	3511	0.442	1.18	1,00E-10	2.3e-09	1.49e-09
BP	GO:1901576	organic substance biosynthetic process	3549	0.44	1.18	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009058	biosynthetic process	3570	0.439	1.17	1,00E-10	2.3e-09	1.49e-09
BP	GO:0050896	response to stimulus	3766	0.511	1.37	1,00E-10	2.3e-09	1.49e-09
BP	GO:0043170	macromolecule metabolic process	4260	0.472	1.26	1,00E-10	2.3e-09	1.49e-09
BP	GO:0006807	nitrogen compound metabolic process	4459	0.463	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044237	cellular metabolic process	4552	0.462	1.24	1,00E-10	2.3e-09	1.49e-09
BP	GO:0044238	primary metabolic process	4640	0.457	1.22	1,00E-10	2.3e-09	1.49e-09
BP	GO:0050794	regulation of cellular process	4668	0.506	1.35	1,00E-10	2.3e-09	1.49e-09
BP	GO:0050789	regulation of biological process	4887	0.514	1.38	1,00E-10	2.3e-09	1.49e-09
BP	GO:0071704	organic substance metabolic process	4916	0.466	1.25	1,00E-10	2.3e-09	1.49e-09
BP	GO:0065007	biological regulation	5036	0.508	1.36	1,00E-10	2.3e-09	1.49e-09
BP	GO:0008152	metabolic process	5141	0.466	1.25	1,00E-10	2.3e-09	1.49e-09
BP	GO:0009987	cellular process	6840	0.527	1.41	1,00E-10	2.3e-09	1.49e-09
BP	GO:0008150	biological_process	7195	0.538	1.44	1,00E-10	2.3e-09	1.49e-09
CC	GO:0031672	A band	10	0.762	1.61	0.00755	0.0311	0.0202
CC	GO:0034358	plasma lipoprotein particle	13	0.744	1.62	0.00324	0.0155	0.01
CC	GO:1990777	lipoprotein particle	13	0.744	1.62	0.00324	0.0155	0.01
CC	GO:0012507	ER to Golgi transport vesicle membrane	34	0.649	1.53	0.00194	0.0101	0.00655
CC	GO:0022625	cytosolic large ribosomal subunit	52	0.565	1.36	0.00554	0.0242	0.0157
CC	GO:1905369	endopeptidase complex	66	0.684	1.67	3,00E-08	4.65e-07	3.01e-07
CC	GO:1902911	protein kinase complex	77	0.532	1.31	0.00591	0.0255	0.0165

CC	GO:0000786	nucleosome	79	0.76	1.88	1,00E-10	2.3e-09	1.49e-09
CC	GO:0072686	mitotic spindle	98	0.715	1.79	1,00E-10	2.3e-09	1.49e-09
CC	GO:0031461	cullin-RING ubiquitin ligase complex	99	0.508	1.27	0.00919	0.0366	0.0237
CC	GO:0000775	chromosome	centromeric region	182	0.615	1.58	1,00E-10	2.3e-09
CC	GO:0043025	neuronal cell body	186	0.504	1.29	0.000212	0.00153	0.000995
CC	GO:0005819	spindle	231	0.611	1.58	1,00E-10	2.3e-09	1.49e-09
CC	GO:0098687	chromosomal region	260	0.605	1.57	1,00E-10	2.3e-09	1.49e-09
CC	GO:0015630	microtubule cytoskeleton	612	0.519	1.37	1,00E-10	2.3e-09	1.49e-09
CC	GO:0042175	nuclear outer membrane-endoplasmic reticulum membrane network	612	0.426	1.12	0.00613	0.0262	0.017
CC	GO:0000785	chromatin	613	0.56	1.47	1,00E-10	2.3e-09	1.49e-09
CC	GO:0032993	protein-DNA complex	649	0.557	1.47	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005694	chromosome	953	0.559	1.48	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005856	cytoskeleton	1011	0.487	1.29	1,00E-10	2.3e-09	1.49e-09
CC	GO:0070062	extracellular exosome	1220	0.501	1.33	1,00E-10	2.3e-09	1.49e-09
CC	GO:1903561	extracellular vesicle	1227	0.5	1.33	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043230	extracellular organelle	1228	0.5	1.33	1,00E-10	2.3e-09	1.49e-09
CC	GO:0065010	extracellular membrane-bounded organelle	1228	0.5	1.33	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005615	extracellular space	1559	0.513	1.37	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005576	extracellular region	1806	0.515	1.37	1,00E-10	2.3e-09	1.49e-09
CC	GO:0031982	vesicle	2102	0.483	1.29	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005886	plasma membrane	2207	0.461	1.23	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005654	nucleoplasm	2262	0.498	1.33	1,00E-10	2.3e-09	1.49e-09
CC	GO:0071944	cell periphery	2391	0.473	1.26	1,00E-10	2.3e-09	1.49e-09
CC	GO:0031981	nuclear lumen	2451	0.493	1.32	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043232	intracellular non-membrane-bounded organelle	2500	0.508	1.35	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043228	non-membrane-bounded organelle	2501	0.507	1.35	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005829	cytosol	2937	0.468	1.25	1,00E-10	2.3e-09	1.49e-09

CC	GO:0031974	membrane-enclosed lumen	3011	0.492	1.31	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043233	organelle lumen	3011	0.492	1.31	1,00E-10	2.3e-09	1.49e-09
CC	GO:0070013	intracellular organelle lumen	3011	0.492	1.31	1,00E-10	2.3e-09	1.49e-09
CC	GO:0032991	protein-containing complex	3126	0.497	1.33	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005634	nucleus	3633	0.494	1.32	1,00E-10	2.3e-09	1.49e-09
CC	GO:0016020	membrane	4051	0.479	1.28	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005737	cytoplasm	5636	0.493	1.32	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043231	intracellular membrane-bounded organelle	5718	0.491	1.32	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043229	intracellular organelle	6042	0.497	1.33	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043227	membrane-bounded organelle	6044	0.499	1.34	1,00E-10	2.3e-09	1.49e-09
CC	GO:0043226	organelle	6312	0.508	1.36	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005622	intracellular anatomical structure	6678	0.508	1.36	1,00E-10	2.3e-09	1.49e-09
CC	GO:0110165	cellular anatomical entity	7408	0.546	1.46	1,00E-10	2.3e-09	1.49e-09
CC	GO:0005575	cellular component	7451	0.547	1.46	1,00E-10	2.3e-09	1.49e-09
MF	GO:0046965	nuclear retinoid X receptor binding	11	0.753	1.61	0.0074	0.0306	0.0198
MF	GO:0061133	endopeptidase activator activity	13	0.755	1.64	0.00185	0.00967	0.00627
MF	GO:0008009	chemokine activity	21	0.793	1.8	1.33e-05	0.000131	8.52e-05
MF	GO:0031491	nucleosome binding	65	0.687	1.68	6.55e-08	9.49e-07	6.15e-07
MF	GO:0015078	proton transmembrane transporter activity	65	0.558	1.36	0.00465	0.021	0.0136
MF	GO:0030527	structural constituent of chromatin	66	0.773	1.89	1,00E-10	2.3e-09	1.49e-09
MF	GO:0005178	integrin binding	86	0.53	1.32	0.00364	0.0171	0.0111
MF	GO:0140993	histone modifying activity	98	0.527	1.32	0.0035	0.0165	0.0107
MF	GO:0008017	microtubule binding	115	0.637	1.61	3.12e-09	5.59e-08	3.62e-08
MF	GO:0004175	endopeptidase activity	160	0.508	1.3	0.000545	0.00342	0.00221
MF	GO:0046982	protein heterodimerization activity	190	0.618	1.58	1,00E-10	2.3e-09	1.49e-09
MF	GO:0005525	GTP binding	190	0.477	1.22	0.00502	0.0224	0.0145
MF	GO:0140678	molecular function inhibitor activity	232	0.576	1.49	1,00E-10	2.3e-09	1.49e-09
MF	GO:0015075	monoatomic ion transmembrane transporter activity	252	0.452	1.17	0.0104	0.0402	0.0261

MF	GO:0140657	ATP-dependent activity	267	0.491	1.27	0.000105	0.000833	0.00054
MF	GO:0003712	transcription coregulator activity	270	0.475	1.23	0.000536	0.00338	0.00219
MF	GO:0004672	protein kinase activity	282	0.472	1.22	0.000449	0.00291	0.00189
MF	GO:0050839	cell adhesion molecule binding	333	0.55	1.43	1,00E-10	2.3e-09	1.49e-09
MF	GO:0016301	kinase activity	362	0.459	1.2	0.000774	0.00461	0.00299
MF	GO:0016462	pyrophosphatase activity	390	0.48	1.25	8.68e-06	8.9e-05	5.77e-05
MF	GO:0005198	structural molecule activity	403	0.545	1.42	1,00E-10	2.3e-09	1.49e-09
MF	GO:0005215	transporter activity	460	0.451	1.18	0.000737	0.00443	0.00287
MF	GO:0140677	molecular function activator activity	473	0.487	1.28	2.86e-07	3.78e-06	2.45e-06
MF	GO:0046983	protein dimerization activity	549	0.543	1.43	1,00E-10	2.3e-09	1.49e-09
MF	GO:0060090	molecular adaptor activity	599	0.46	1.21	1.8e-05	0.000172	0.000111
MF	GO:0005102	signaling receptor binding	634	0.513	1.35	1,00E-10	2.3e-09	1.49e-09
MF	GO:0140110	transcription regulator activity	727	0.449	1.18	5.08e-05	0.000437	0.000283
MF	GO:0035639	purine ribonucleoside triphosphate binding	895	0.48	1.27	1,00E-10	2.3e-09	1.49e-09
MF	GO:0098772	molecular function regulator activity	903	0.497	1.32	1,00E-10	2.3e-09	1.49e-09
MF	GO:0032555	purine ribonucleotide binding	919	0.479	1.27	1,00E-10	2.3e-09	1.49e-09
MF	GO:0017076	purine nucleotide binding	965	0.477	1.26	1,00E-10	2.3e-09	1.49e-09
MF	GO:0044877	protein-containing complex binding	967	0.505	1.34	1,00E-10	2.3e-09	1.49e-09
MF	GO:0042802	identical protein binding	1035	0.514	1.36	1,00E-10	2.3e-09	1.49e-09
MF	GO:0003677	DNA binding	1047	0.521	1.38	1,00E-10	2.3e-09	1.49e-09
MF	GO:0097367	carbohydrate derivative binding	1071	0.486	1.29	1,00E-10	2.3e-09	1.49e-09
MF	GO:0019899	enzyme binding	1153	0.494	1.31	1,00E-10	2.3e-09	1.49e-09
MF	GO:0003676	nucleic acid binding	1908	0.483	1.29	1,00E-10	2.3e-09	1.49e-09
MF	GO:0003824	catalytic activity	2585	0.425	1.13	4.05e-07	5.24e-06	3.4e-06
MF	GO:0097159	organic cyclic compound binding	2804	0.476	1.27	1,00E-10	2.3e-09	1.49e-09
MF	GO:0005515	protein binding	6290	0.531	1.42	1,00E-10	2.3e-09	1.49e-09
MF	GO:0005488	binding	6904	0.538	1.44	1,00E-10	2.3e-09	1.49e-09
MF	GO:0003674	molecular_function	7325	0.546	1.46	1,00E-10	2.3e-09	1.49e-09
Pathways enriched in genes lower expressed in winter-spring season								

ONTOLOGY	ID	Description	setSize	enrichmentScore	NES	pvalue	p.adjust	qvalue
BP	GO:1901016	regulation of potassium ion transmembrane transporter activity	18	-0.732	-1.53	0.000774	0.0336	0.0331
BP	GO:0007288	sperm axoneme assembly	23	-0.738	-1.57	0.000162	0.00888	0.00875
BP	GO:0003352	regulation of cilium movement	24	-0.716	-1.54	0.000196	0.0104	0.0102
BP	GO:0031016	pancreas development	24	-0.683	-1.47	0.00108	0.0446	0.044
BP	GO:0120316	sperm flagellum assembly	33	-0.742	-1.63	5.29e-06	0.000442	0.000436
BP	GO:0006858	extracellular transport	42	-0.714	-1.6	1.53e-06	0.000157	0.000154
BP	GO:0042073	intraciliary transport	44	-0.699	-1.57	3.77e-06	0.000347	0.000342
BP	GO:0061512	protein localization to cilium	49	-0.688	-1.55	2.7e-06	0.000259	0.000255
BP	GO:0007368	determination of left/right symmetry	67	-0.637	-1.46	6.67e-06	0.000539	0.000531
BP	GO:0007224	smoothened signaling pathway	73	-0.582	-1.34	0.000481	0.0228	0.0225
BP	GO:0030317	flagellated sperm motility	76	-0.729	-1.69	1,00E-10	1.59e-08	1.56e-08
BP	GO:0097722	sperm motility	76	-0.729	-1.69	1,00E-10	1.59e-08	1.56e-08
BP	GO:0031503	protein-containing complex localization	84	-0.639	-1.49	6.43e-07	7.59e-05	7.47e-05
BP	GO:0048515	spermatid differentiation	94	-0.657	-1.53	7.3e-09	9.88e-07	9.73e-07
BP	GO:0060294	cilium movement involved in cell motility	97	-0.713	-1.67	1,00E-10	1.59e-08	1.56e-08
BP	GO:0001539	cilium or flagellum-dependent cell motility	102	-0.706	-1.65	1,00E-10	1.59e-08	1.56e-08
BP	GO:0060285	cilium-dependent cell motility	102	-0.706	-1.65	1,00E-10	1.59e-08	1.56e-08
BP	GO:0032886	regulation of microtubule-based process	111	-0.603	-1.42	9.5e-07	0.000102	1,00E-04
BP	GO:0003341	cilium movement	134	-0.706	-1.67	1,00E-10	1.59e-08	1.56e-08
BP	GO:0099111	microtubule-based transport	134	-0.645	-1.52	1,00E-10	1.59e-08	1.56e-08
BP	GO:0022412	cellular process involved in reproduction in multicellular organism	171	-0.56	-1.33	3.97e-06	0.000359	0.000353
BP	GO:0007018	microtubule-based movement	250	-0.656	-1.57	1,00E-10	1.59e-08	1.56e-08
BP	GO:0030001	metal ion transport	250	-0.507	-1.22	0.000536	0.0252	0.0248
BP	GO:0060271	cilium assembly	273	-0.642	-1.55	1,00E-10	1.59e-08	1.56e-08
BP	GO:0044782	cilium organization	290	-0.64	-1.54	1,00E-10	1.59e-08	1.56e-08
BP	GO:0000226	microtubule cytoskeleton organization	315	-0.577	-1.39	1,00E-10	1.59e-08	1.56e-08

BP	GO:0120031	plasma membrane bounded cell projection assembly	335	-0.617	-1.49	1,00E-10	1.59e-08	1.56e-08
BP	GO:0030031	cell projection assembly	339	-0.616	-1.49	1,00E-10	1.59e-08	1.56e-08
BP	GO:0022414	reproductive process	469	-0.494	-1.2	1.23e-05	0.000898	0.000885
BP	GO:0007017	microtubule-based process	471	-0.603	-1.47	1,00E-10	1.59e-08	1.56e-08
BP	GO:0070925	organelle assembly	475	-0.571	-1.39	1,00E-10	1.59e-08	1.56e-08
BP	GO:0007010	cytoskeleton organization	588	-0.524	-1.28	1,00E-10	1.59e-08	1.56e-08
BP	GO:0120036	plasma membrane bounded cell projection organization	657	-0.542	-1.33	1,00E-10	1.59e-08	1.56e-08
BP	GO:0030030	cell projection organization	673	-0.545	-1.33	1,00E-10	1.59e-08	1.56e-08
CC	GO:0090543	Flemming body	14	-0.777	-1.59	0.000444	0.0218	0.0214
CC	GO:0030315	T-tubule	24	-0.698	-1.5	0.000463	0.0222	0.0218
CC	GO:0030990	intraciliary transport particle	25	-0.7	-1.51	0.000847	0.0364	0.0359
CC	GO:0034703	cation channel complex	64	-0.596	-1.37	0.000399	0.0199	0.0197
CC	GO:0097729	9+2 motile cilium	107	-0.692	-1.62	1,00E-10	1.59e-08	1.56e-08
CC	GO:0005930	axoneme	144	-0.657	-1.55	1,00E-10	1.59e-08	1.56e-08
CC	GO:0097014	ciliary plasm	145	-0.653	-1.55	1,00E-10	1.59e-08	1.56e-08
CC	GO:0031514	motile cilium	165	-0.681	-1.62	1,00E-10	1.59e-08	1.56e-08
CC	GO:0032838	plasma membrane bounded cell projection cytoplasm	180	-0.621	-1.48	1,00E-10	1.59e-08	1.56e-08
CC	GO:0099081	supramolecular polymer	361	-0.501	-1.21	1.69e-05	0.00121	0.0012
CC	GO:0005815	microtubule organizing center	440	-0.547	-1.33	1,00E-10	1.59e-08	1.56e-08
CC	GO:0005929	cilium	447	-0.634	-1.54	1,00E-10	1.59e-08	1.56e-08
CC	GO:0015630	microtubule cytoskeleton	661	-0.546	-1.33	1,00E-10	1.59e-08	1.56e-08
CC	GO:0120025	plasma membrane bounded cell projection	917	-0.545	-1.34	1,00E-10	1.59e-08	1.56e-08
CC	GO:0042995	cell projection	955	-0.548	-1.35	1,00E-10	1.59e-08	1.56e-08
CC	GO:0005856	cytoskeleton	972	-0.538	-1.32	1,00E-10	1.59e-08	1.56e-08
CC	GO:0043228	non-membrane-bounded organelle	1859	-0.443	-1.09	0.000572	0.0263	0.026
MF	GO:0008569	minus-end-directed microtubule motor activity	13	-0.876	-1.78	4.87e-06	0.000423	0.000417
MF	GO:0051959	dynein light intermediate chain binding	19	-0.805	-1.69	9.52e-06	0.000717	0.000706

MF	GO:0045505	dynein intermediate chain binding	26	-0.78	-1.69	2.15e-06	0.000215	0.000212
MF	GO:0022843	voltage-gated monoatomic cation channel activity	28	-0.683	-1.48	0.000655	0.0293	0.0288
MF	GO:0003777	microtubule motor activity	37	-0.669	-1.48	0.000317	0.016	0.0158
MF	GO:0008092	cytoskeletal protein binding	373	-0.513	-1.24	3.42e-06	0.000321	0.000317

Supplementary Table 2. *Gene Set Enrichment Analysis (GSEA) of Gene Ontology (GO) terms using the gseGO function from the clusterProfiler package. Genes higher expressed, logFC > 0, and lower expressed, logFC < 0, in winter-spring season were analyzed separately. Genes were ranked using the following metrics: $-\log_{10}$(P-value). simplify function was used to remove redundant GO terms. Terms with FDR < 0.05 were considered significantly enriched*